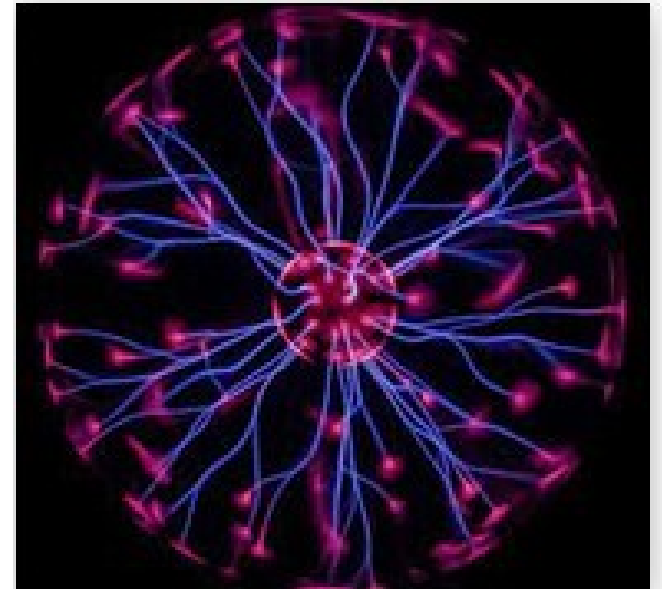
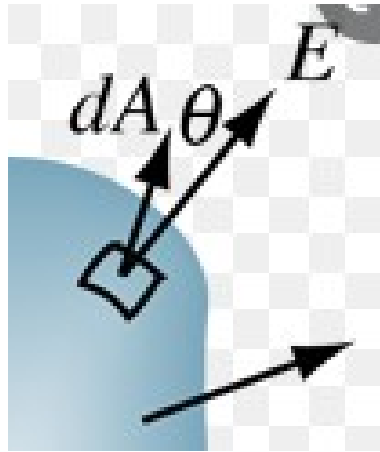


Unit 1.2 Gauss Law

- Field lines and Gauss law
- How to use Gauss law
- the calculus part
- Faradays cage

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$



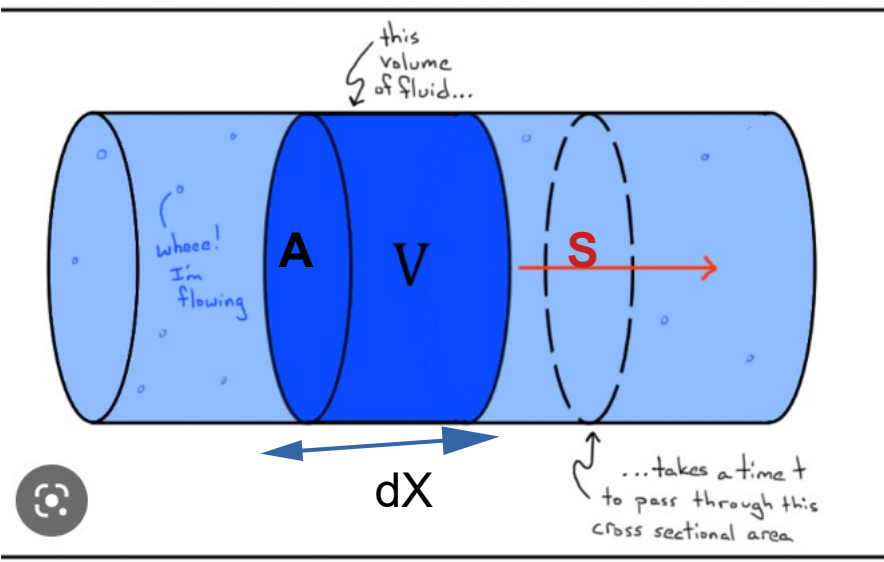
Discuss the picture



Johann Carl Friedrich Gauss

April 1777 – 23 February 1855

- ❑ Flux in Physics is used to two distinct ways.
- ❑ The first meaning is the rate of flow, such as the amount of water flowing in a river, i.e. volume per unit area per unit time. Or, for light, it is the amount of energy per unit area per unit time.

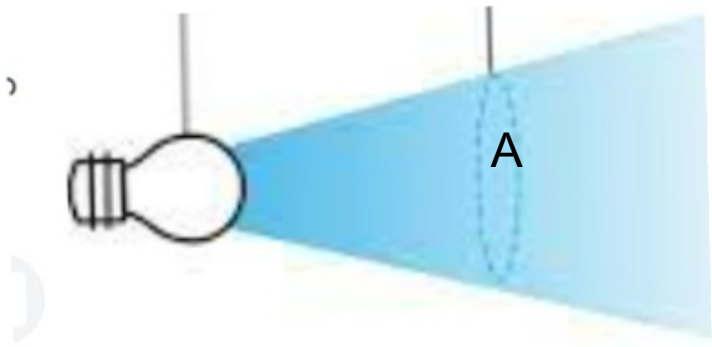


A is area
 V dolume
 S speed
 .dX distance covered in dt by water

See video about flux (canvas)
 Reference: mechanical universe

Flux could be the amount of water
 Going through an open area
 $Volume = A \ dX = A \ S \ dt$

$Volume/dt = \text{area} \times \text{speed}$

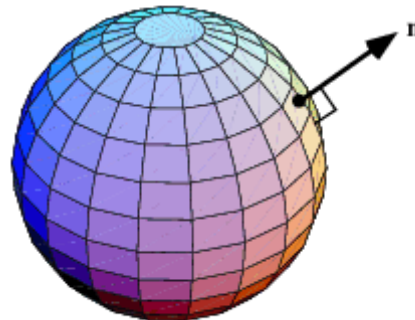
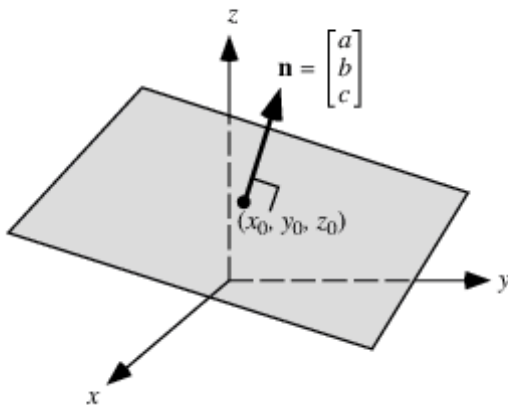
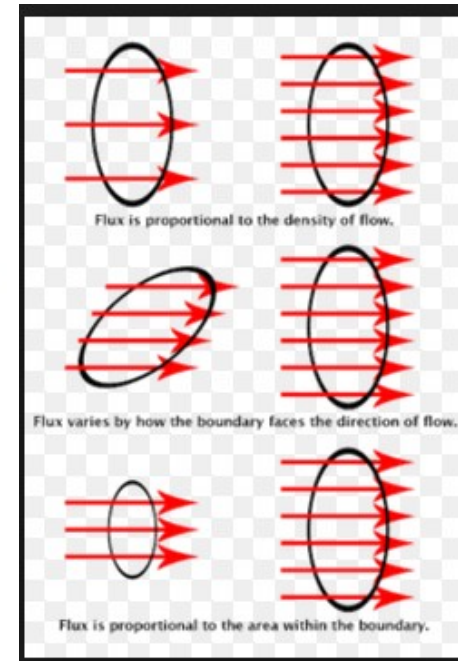
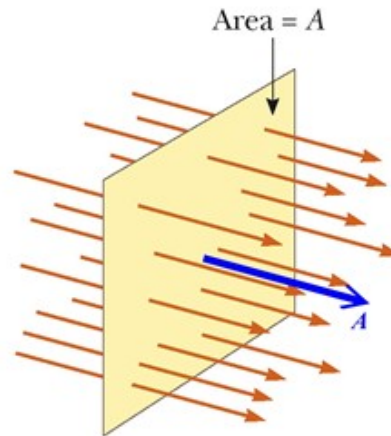


Some calculus → flux of a vector field - continued

FLUX of E is proportional to the total
Number of lines through a surface.

Count the number
Of lines through surface.

**There is no such things
As field lines but it helps
Us visualize the magnitude of E**



- Flux in Physics is used to two distinct ways.
- The first meaning is the rate of flow, such as the amount of water flowing in a river, i.e. volume per unit area per unit time. Or, for light, it is the amount of energy per unit area per unit time.

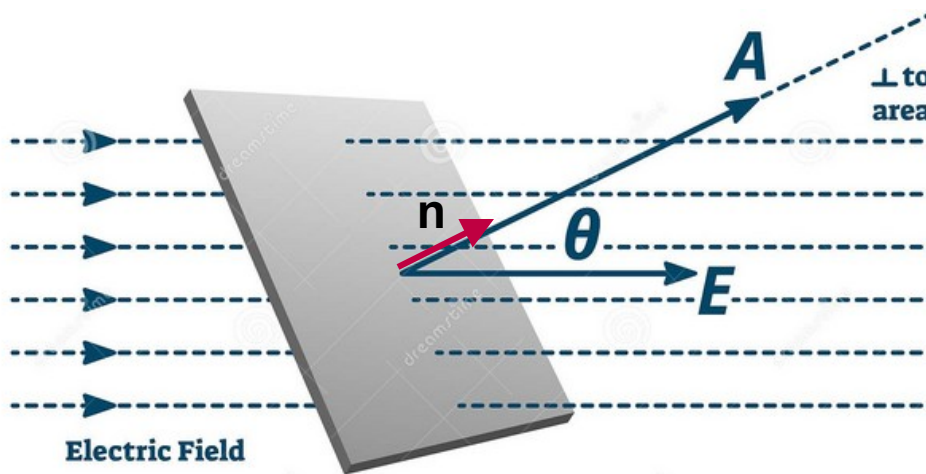
Video flux. And next slide and do example 22.1

If E is uniform the flux of E through an area A

Is $E \cdot A = |E| |A| \cos(\text{angle between } A \text{ and } E)$

A is a vector. The unit vector is $\hat{n}$

If the surface is closed $\hat{n}$ is always outward



Flux $\rightarrow$ use only the E Along A .

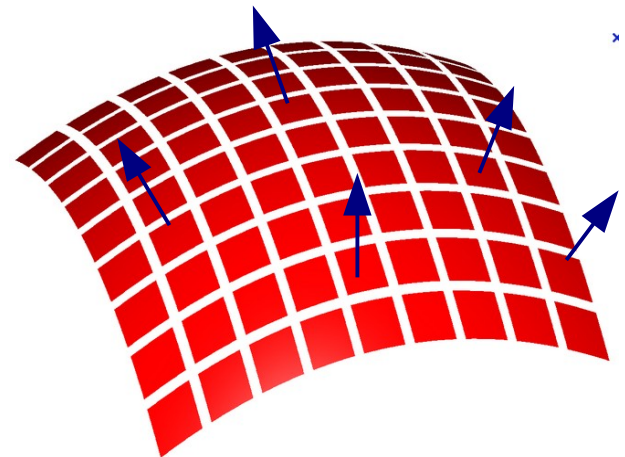
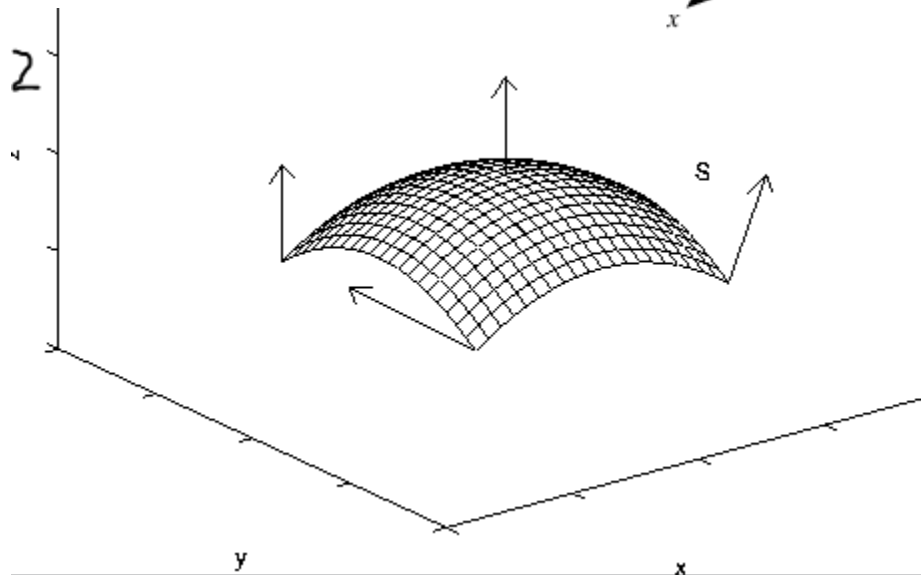
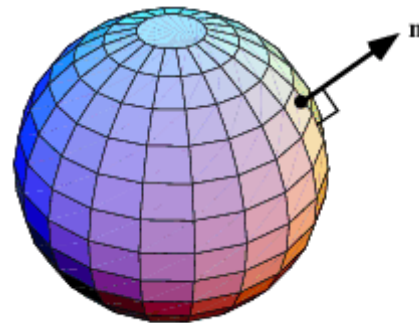
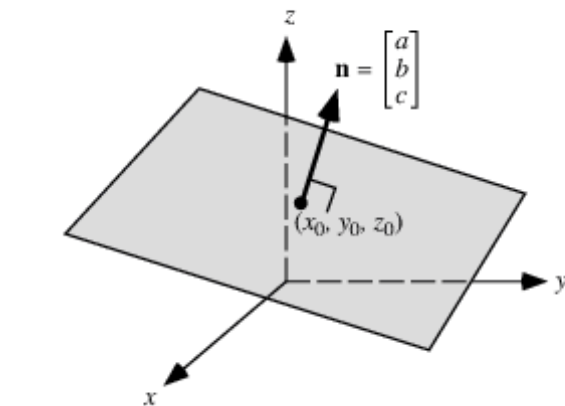
$$\text{Flux} = \phi = EA \cos \theta$$

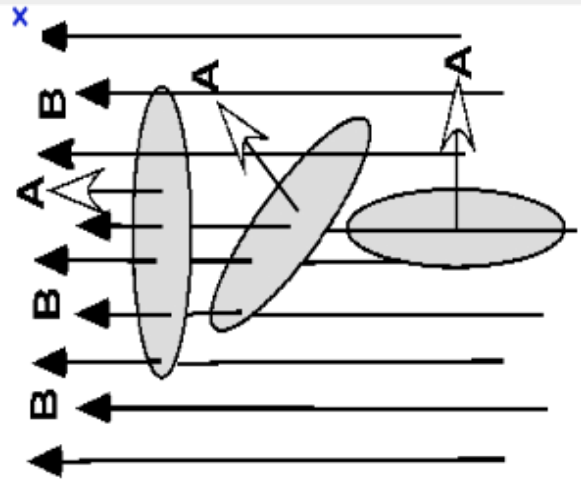
A unit vector is attached to unit areas.

If its not a close surface, the unit vector can be in either direction.

No favorite direction. The unit vector is always perpendicular to the unit area.

For a close surface, the unit vector is always out of the area.

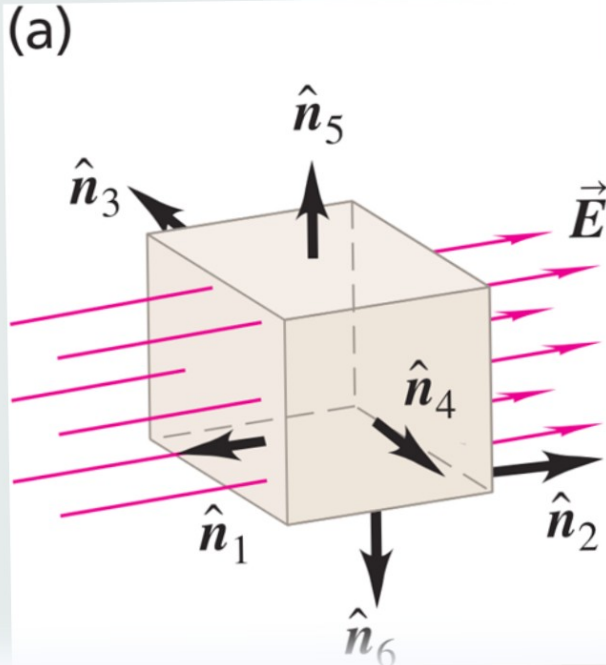




→ The number of lines depends on the angle between the vector field and the area vector

Book problem 22.2

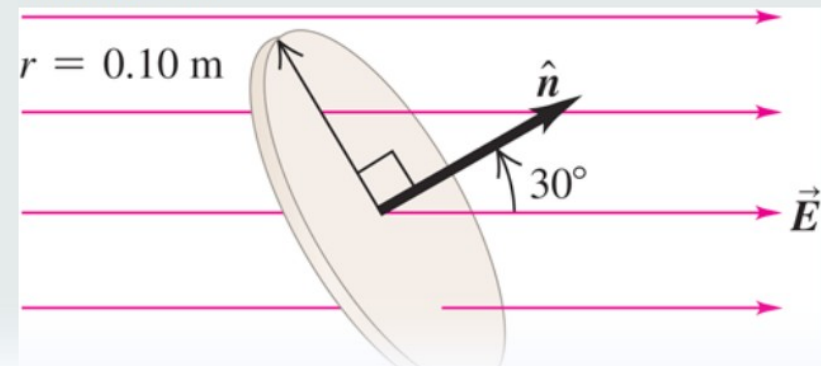
FIGURE 22.8



22.1 Electric flux through a disk

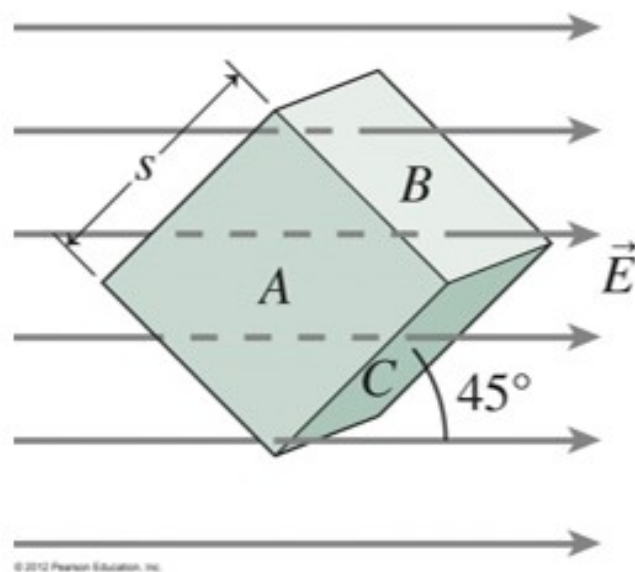
A disk of radius 0.10 m is oriented with its normal unit vector $\hat{n}$ at 30° to a uniform electric field $\vec{E}$ of magnitude $2.0 \times 10^3 \text{ N/C}$ (Fig. 22.7). (Since this isn't a closed surface, it has no "inside" or "outside." That's why we have to specify the direction of $\hat{n}$ in the figure.) (a) What is the electric flux through the disk? (b) What is the flux through the disk if it is turned so that $\hat{n}$ is perpendicular to $\vec{E}$? (c) What is the flux through the disk if $\hat{n}$ is parallel to $\vec{E}$?

FIGURE 22.7



- The flux through side B of the cube in the figure is the same as the flux through side C . What is a correct expression for the flux through each of these sides?

- A. $s^2 E$
- B. $s^3 E$
- C. $s^2 E \cos 45^\circ$
- D. $s^3 E \cos 45^\circ$



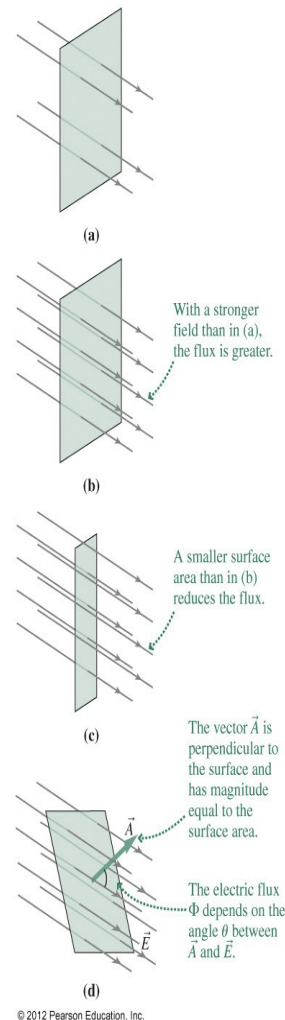
$$\Phi_E = \sum (E \cos \phi) \Delta A$$

Electric Flux

- **Electric flux** quantifies the notion “number of field lines crossing a surface.”
- The electric flux Φ through a flat surface in a uniform electric field depends on the field strength E , the surface area A , and the angle θ between the field and the normal to the surface.
- Mathematically, the flux is given by

$$\Phi = EA \cos \theta = \vec{E} \cdot \vec{A}$$

- Here $\vec{A}$ is a vector whose magnitude is the surface area A and whose orientation is normal to the surface.



Here the surface is like
A paper.

A is a vector
perpendicular to
The surface
Why the dot product?

Electric Flux with Curved Surfaces and Nonuniform Fields

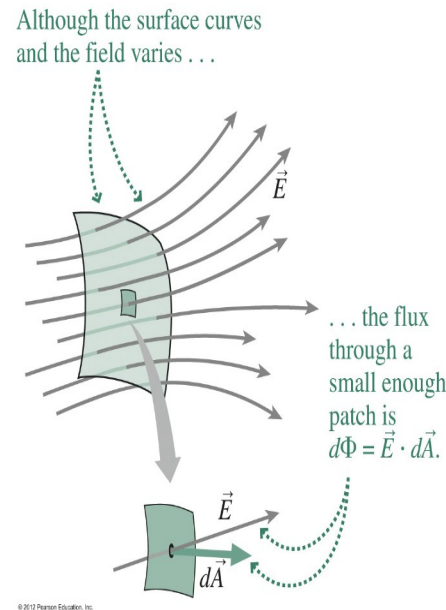
- When the surface is curved or the field is nonuniform, we calculate the flux by dividing the surface into small patches $d\vec{A}$, so small that each patch is essentially flat and the field is essentially uniform over each.

- We then sum the fluxes

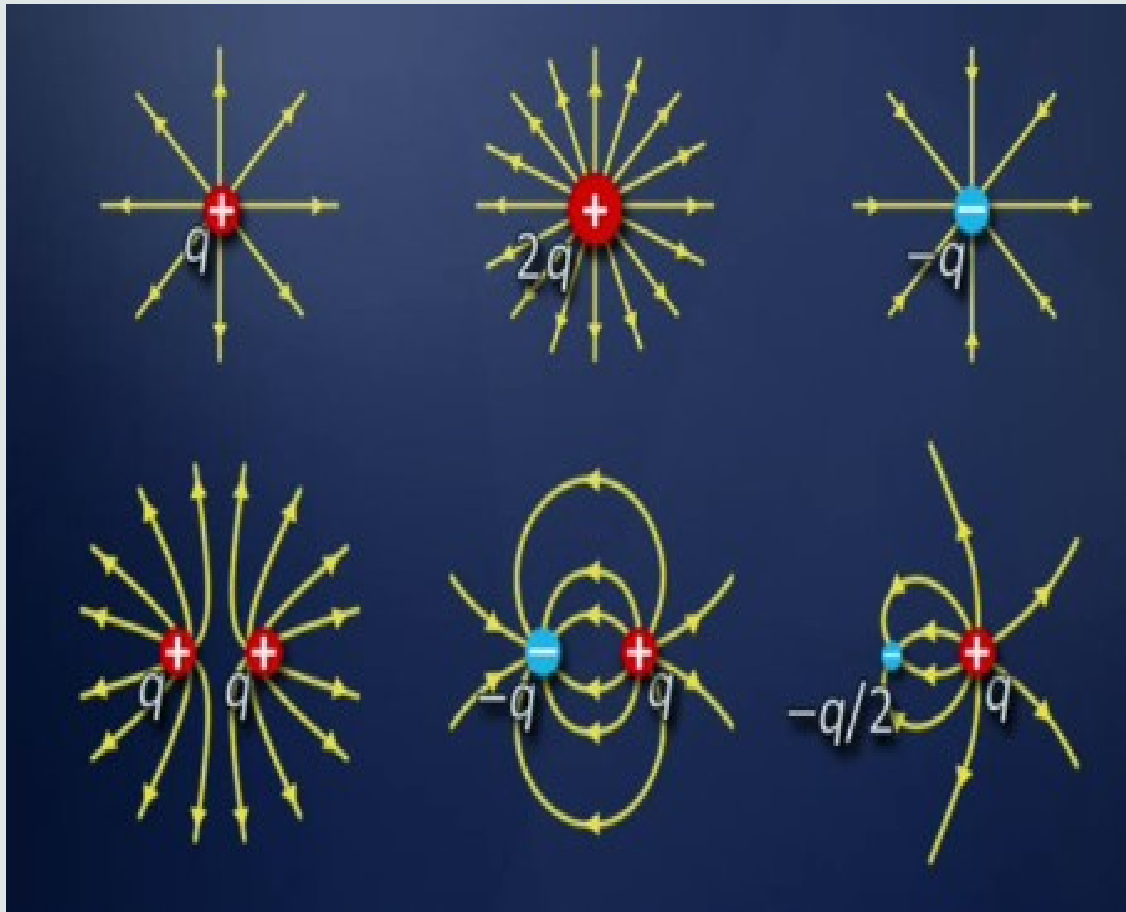
$$d\Phi = \vec{E} \cdot d\vec{A} \text{ over each patch.}$$

- In the limit of infinitely many infinitesimally small patches, the sum becomes a **surface integral**:

$$\Phi = \int \vec{E} \cdot d\vec{A}$$



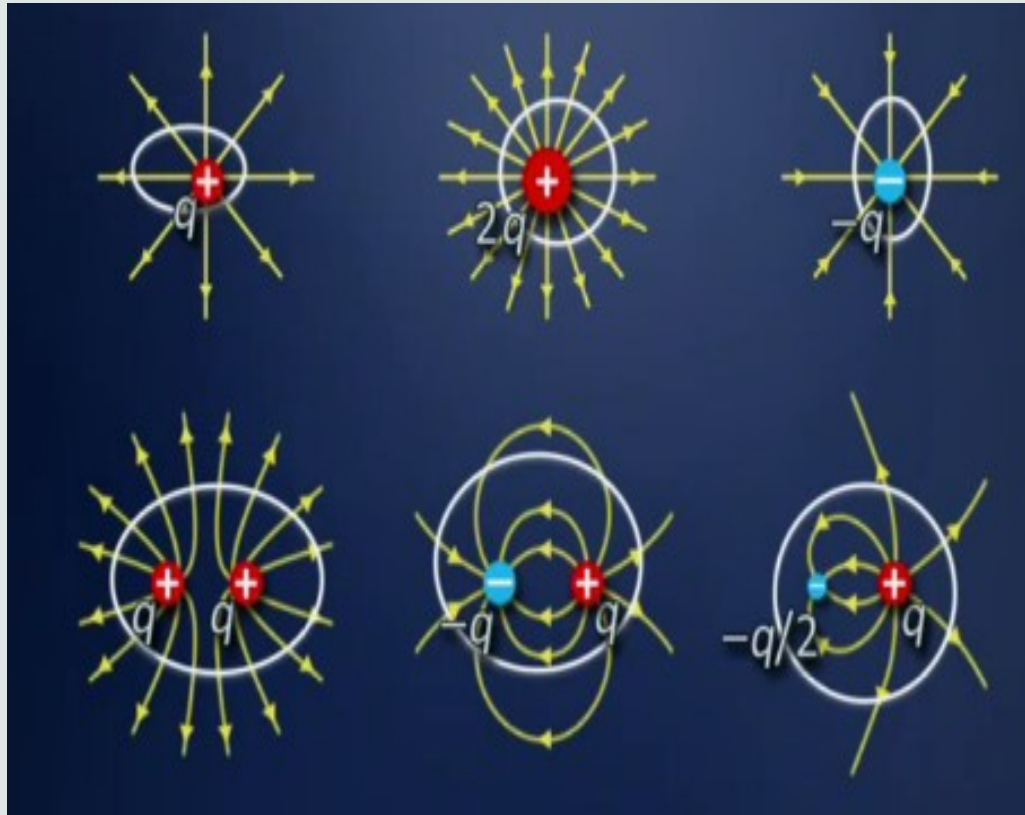
Gauss' law



The number of field lines
Depend on the charge

Let's take a close bag
And let's place the charge(s)
In this bag and let's count
The number of lines
Poking through.

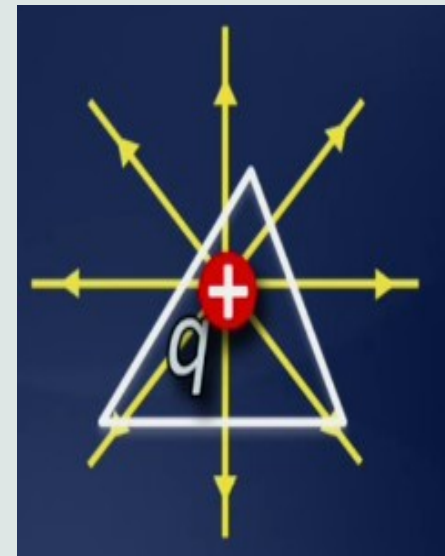
The number of lines
Is proportional to flux
If poking out is positive
Or poking in is negative.



→ Count the number of lines
Getting in (-) or out (+) the bag

→ we see the number of lines
Is proportional to the total charges
In the bag

→ the shape of the bag does not matter



Gauss's Law

- Number of field lines emerging from *any* closed surface depends only on the *enclosed charge*

Video gauss law



This area is not closed Its just a membrane.
Flux = proportional to
number of lines going through a closed surface.
The closed surface is like a bag.

- Mathematically: $\oint_{\text{closed surface}} \vec{E} \cdot d\vec{A} = 4\pi k q_{\text{enclosed}}$

With $k = 1/(4\pi \epsilon_0)$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

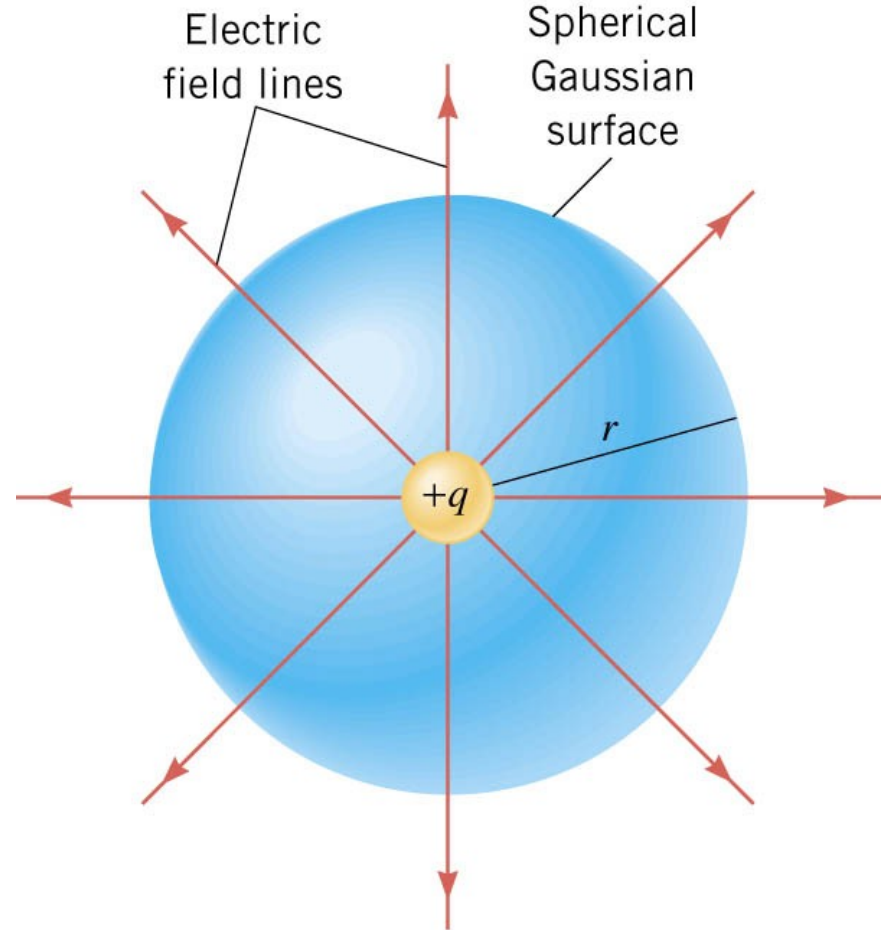
18.9 Gauss' Law

$$E = kq/r^2 = q/(4\pi\epsilon_0 r^2)$$

$$E = q/(A\epsilon_0)$$

$$\underbrace{EA} = \frac{q}{\epsilon_0}$$

Electric flux, $\Phi_E = EA$



This is correct if E is uniform over the close Surface. The blue sphere is a spherical bag. The field lines are poking through.

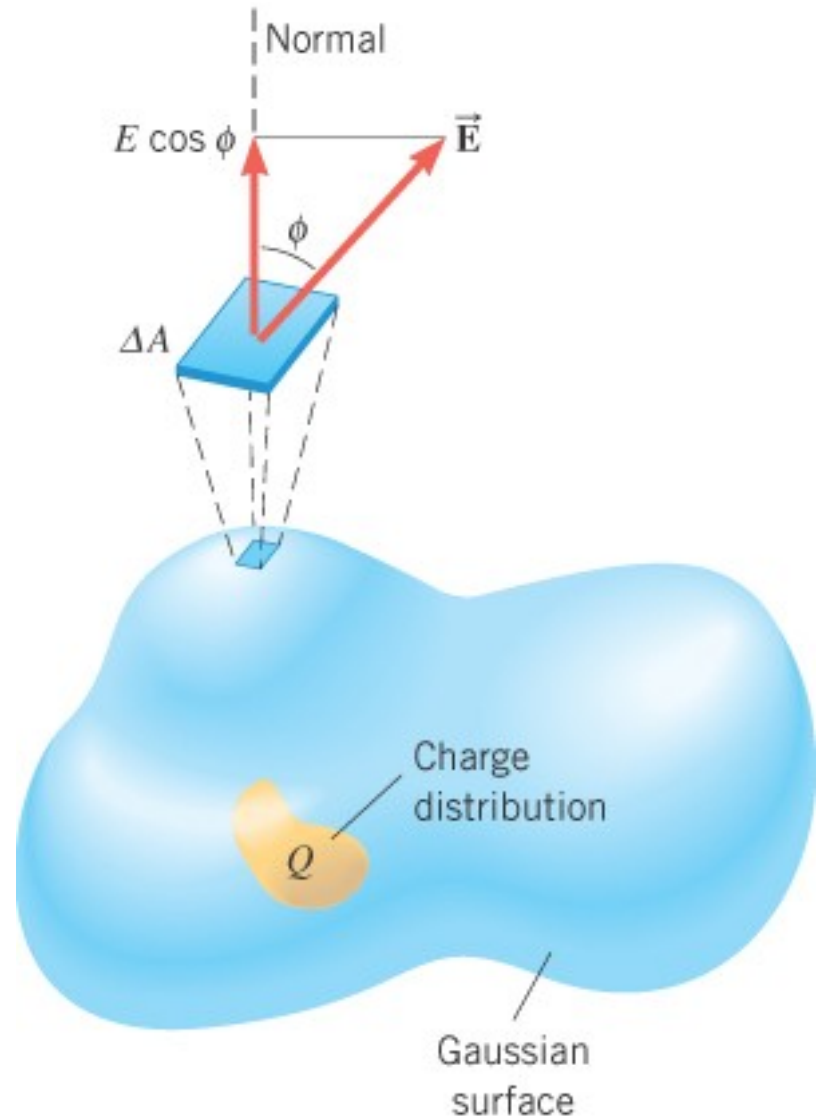
$$\epsilon_0 = 8.85418782 \times 10^{-12} \text{ m}^{-3} \text{ kg}^{-1} \text{ s}^4 \text{ A}^2$$

18.9 Gauss' Law

$$\Phi_E = \sum (E \cos \phi) \Delta A$$

Example 22.1 and do
22.2 on your own.

Video E field



18.9 Gauss' Law

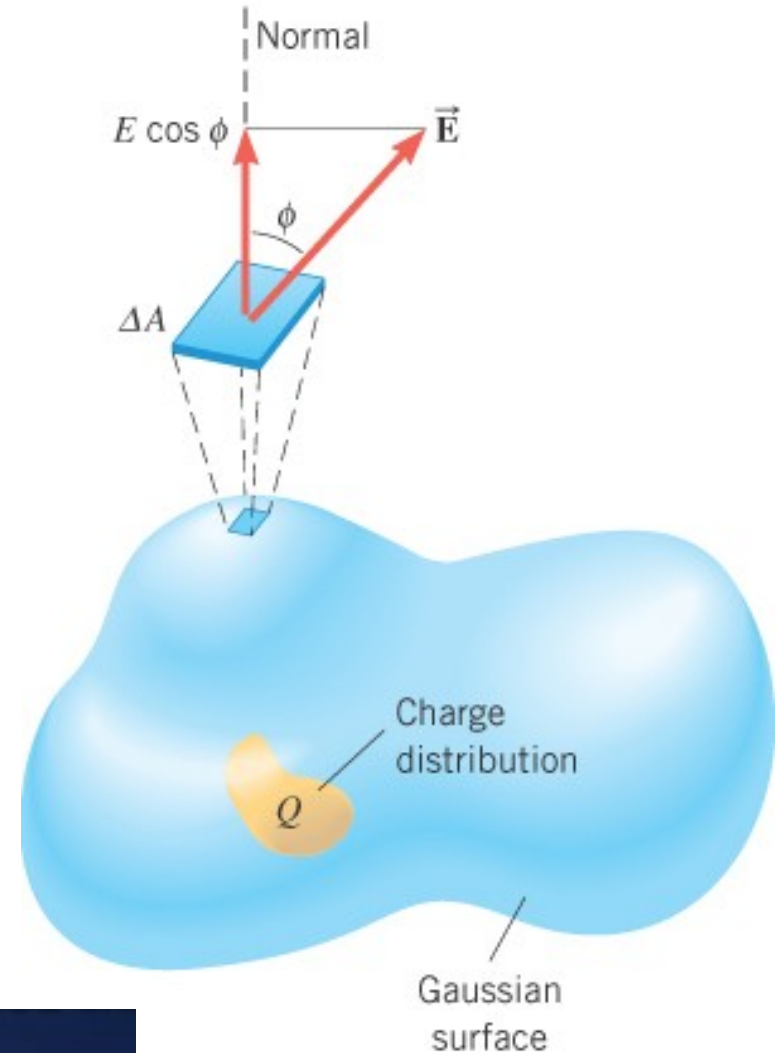
GAUSS' LAW

The electric flux through a Gaussian surface is equal to the net charge enclosed in that surface divided by the permittivity of free space:

$$\sum (E \cos \phi) \Delta A = \frac{Q}{\epsilon_0}$$

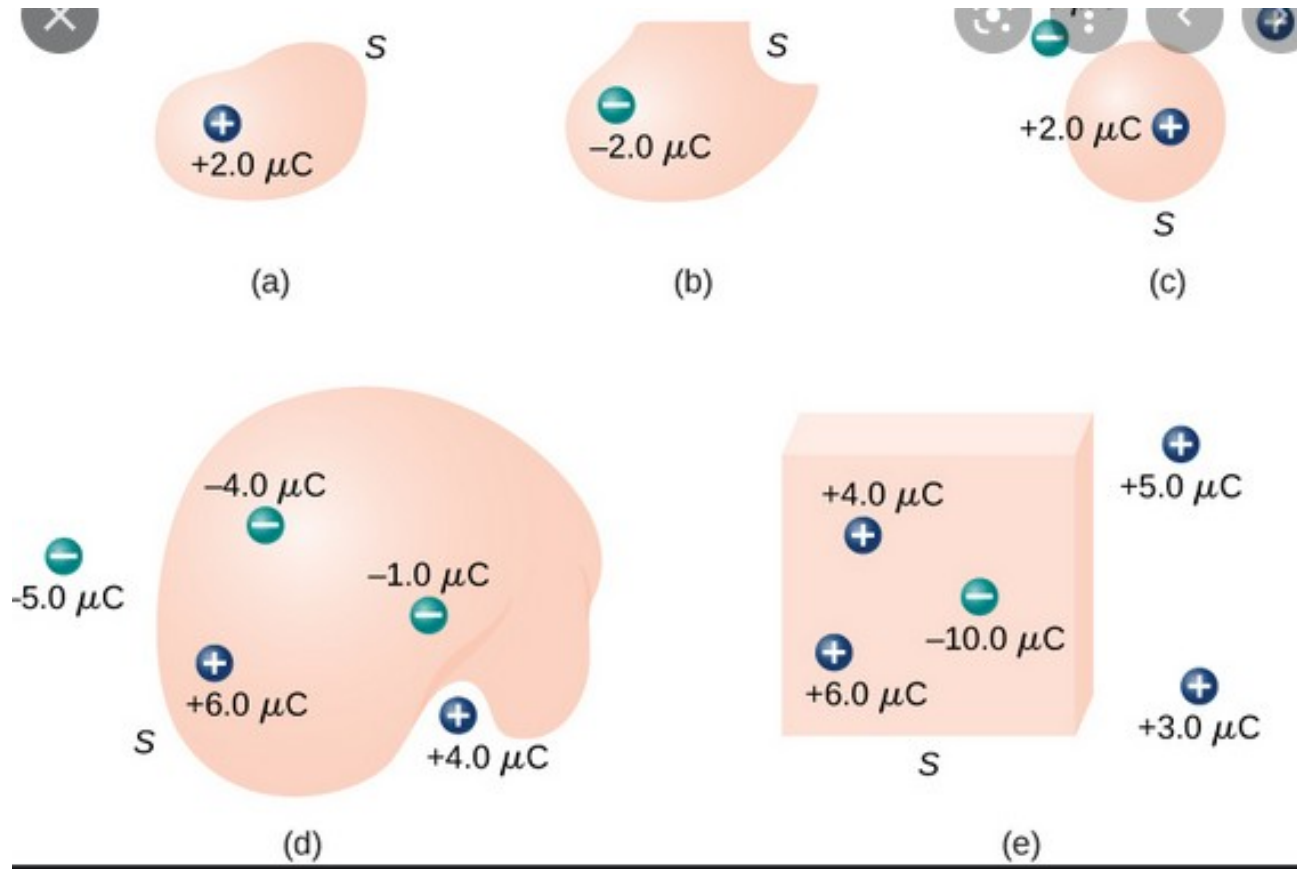
SI Units of Electric Flux: $\text{N} \cdot \text{m}^2/\text{C}$

• Mathematically: $\oint_{\text{closed surface}} \vec{E} \cdot d\vec{A} = 4\pi k q_{\text{enclosed}}$



Flux depends through a surface depends solely on the charge inside the surface.

- A spherical surface surrounds an isolated positive charge. We can calculate the electric flux for this surface. If a second charge is placed outside the spherical surface, what happens to the magnitude of the flux?
 - A. The flux increases proportionally to the magnitude of the second charge.
 - B. The flux decreases proportionally to the magnitude of the second charge.
 - C. The flux does not change.
 - D. The answer depends on whether the second charge is positive or negative.

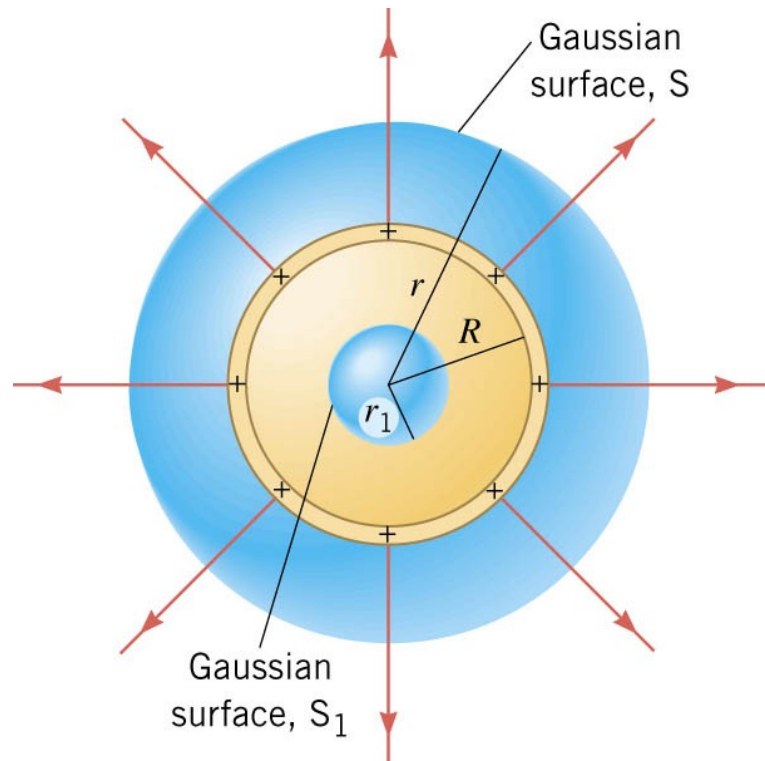


Gaussian surfaces in pink . The flux through the surface only depend on the Charges inside! (Q_{in}/ϵ_0). Try. Positive =outward, negative = inward

Example 15 The Electric Field of a Charged Thin Spherical Shell

A positive charge is spread uniformly over the shell. Find the magnitude of the electric field at any point (a) outside the shell and (b) inside the shell.

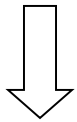
$$\sum (E \cos \phi) \Delta A = \frac{Q}{\epsilon_0}$$



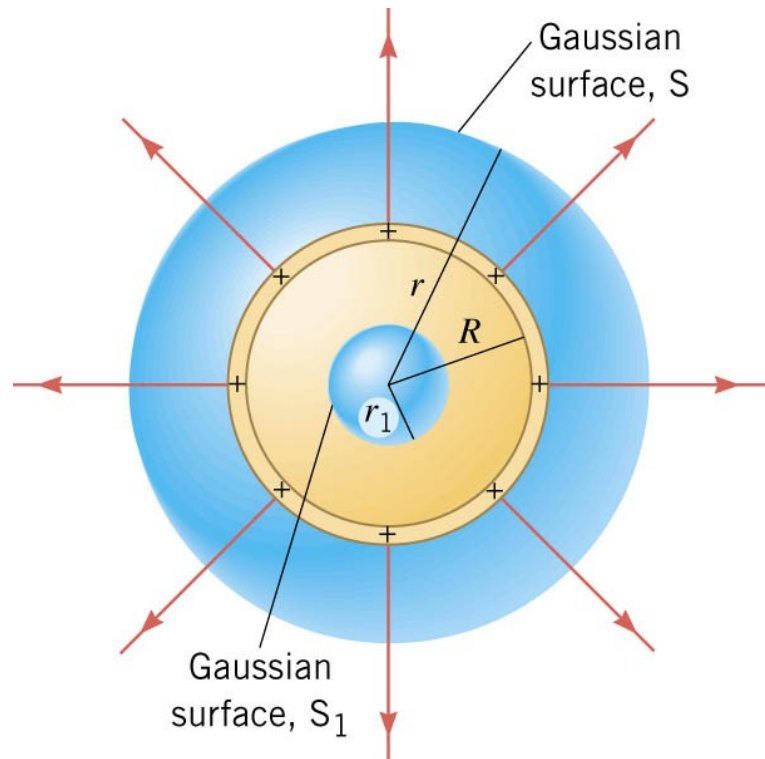
18.9 Gauss' Law

$$\Phi_E = \sum (E \cos \phi) \Delta A = \sum (E \cos 0) \Delta A$$

$$= E \sum \Delta A = E(4\pi r^2)$$



$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$



18.9 Gauss' Law

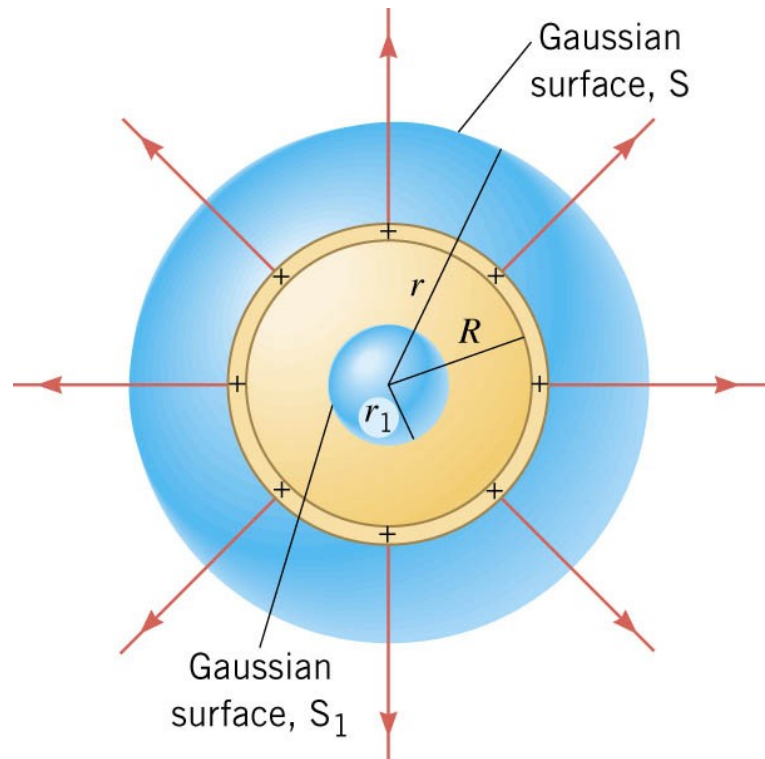
$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$

(a) Outside the shell, the Gaussian surface encloses all of the charge.

$$E = \frac{q}{4\pi r^2 \epsilon_0}$$

(b) Inside the shell, the Gaussian surface encloses no charge.

$$E = 0$$



- A spherical shell of radius R has a total charge $+Q$ uniformly distributed over its surface. The electric field at distance $r > R$ is
 - A. proportional to $1/r$.
 - B. proportional to $1/r^2$.
 - C. proportional to r .
 - D. independent of r .

Make a drawing

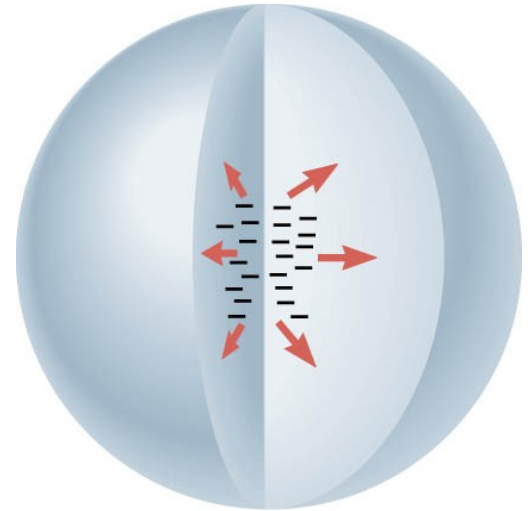
18.8 *The Electric Field Inside a Conductor: Shielding*

Video shielding – Faraday's cage

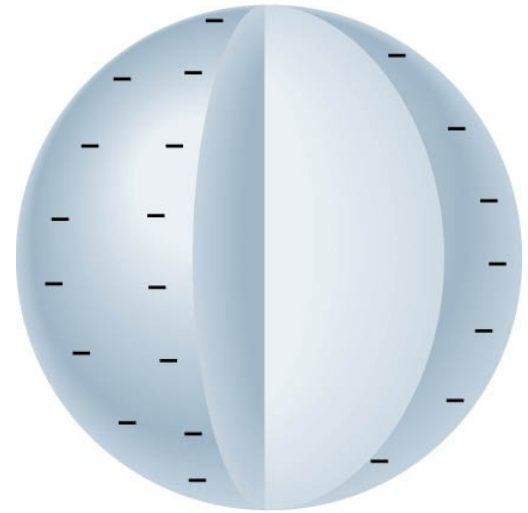
At equilibrium under electrostatic conditions, any excess charge resides on the surface of a conductor.

At equilibrium under electrostatic conditions, the electric field is zero at any point within a conducting material.

The conductor shields any charge within it from electric fields created outside the conductor.



(a)



(b)

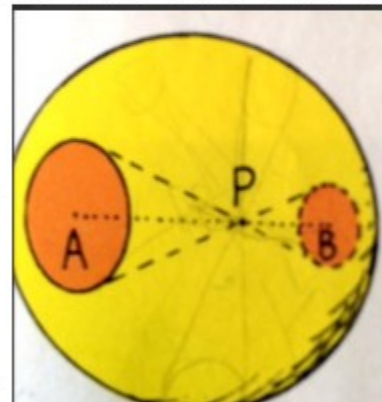
Gauss law is a consequence of inverse square law

Theorem 1 :

A uniform spherical shell of charge behaves, for external points, as if all its charge were concentrated at its center.

Theorem 2 :

A uniform spherical shell of charge exerts no force on a charged particle placed inside the shell.



Source: Paul Hewitt
Conceptual Physics

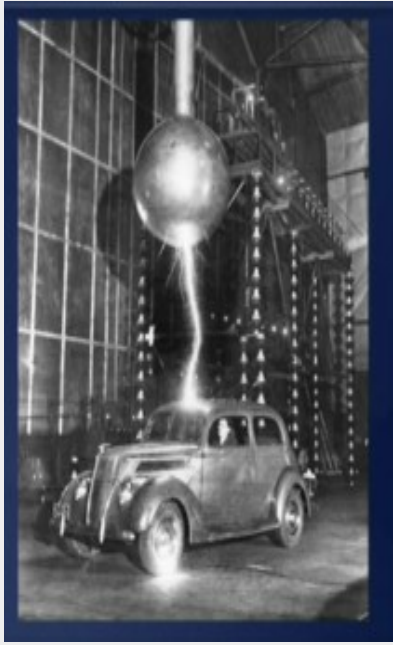
- A spherical shell carries charge Q uniformly distributed over its surface. If the charge on the shell doubles, what happens to the electric field strength inside the shell?
 - A. The electric field strength is zero.
 - B. The electric field strength quadruples.
 - C. The electric field strength is halved.
 - D. The electric field strength doubles.



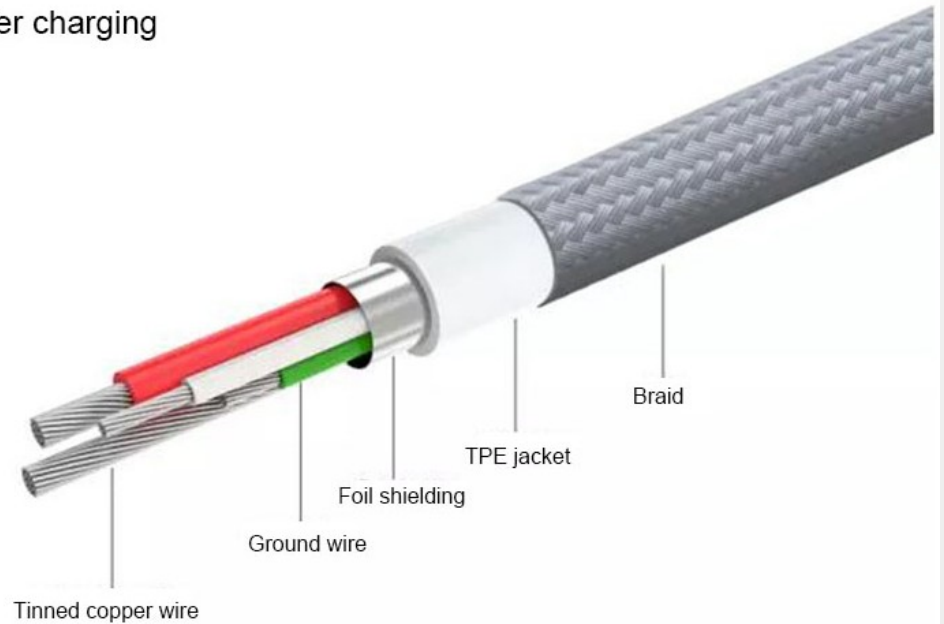
Faradays
cage

→ was used for linear accelerator.
Donated to Boston Museum

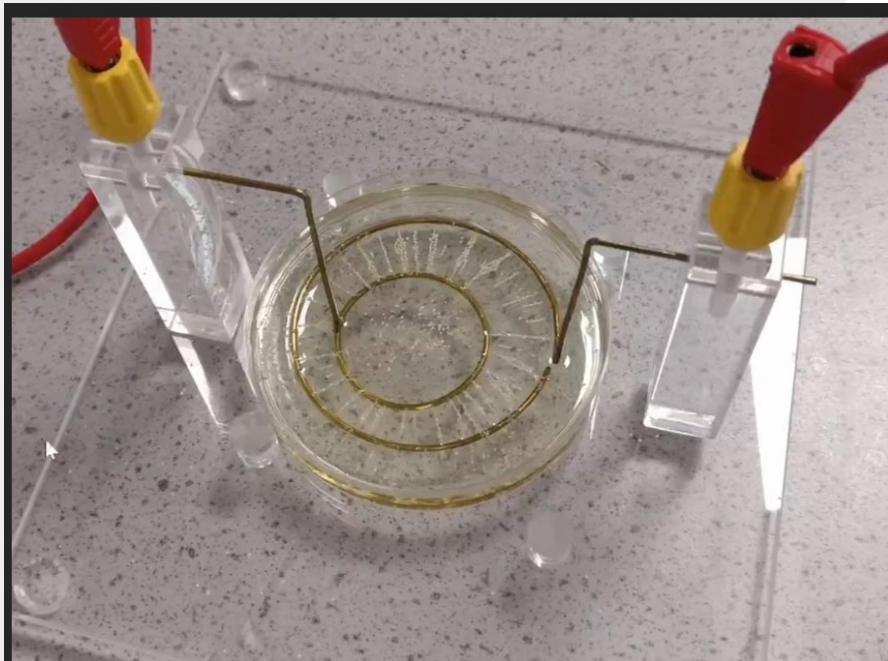
→ Stray charges will move on the outside conductor. The signal is shielded from noise.



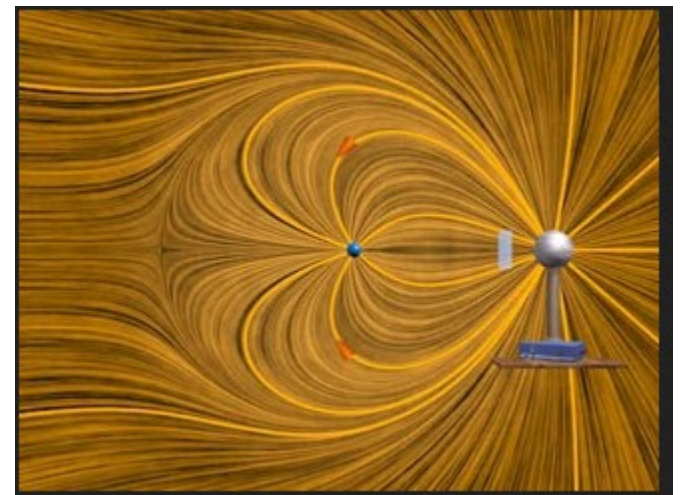
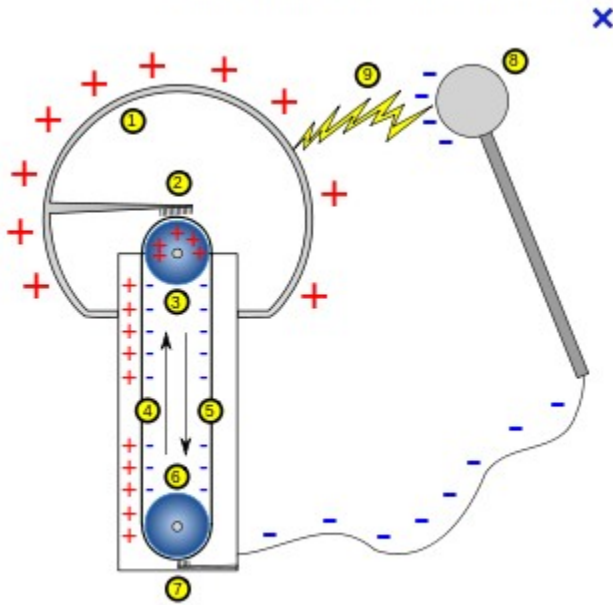
Tinned-copper wire + foil shielding + ground wire lower resistance
Faster transmission
Safer charging



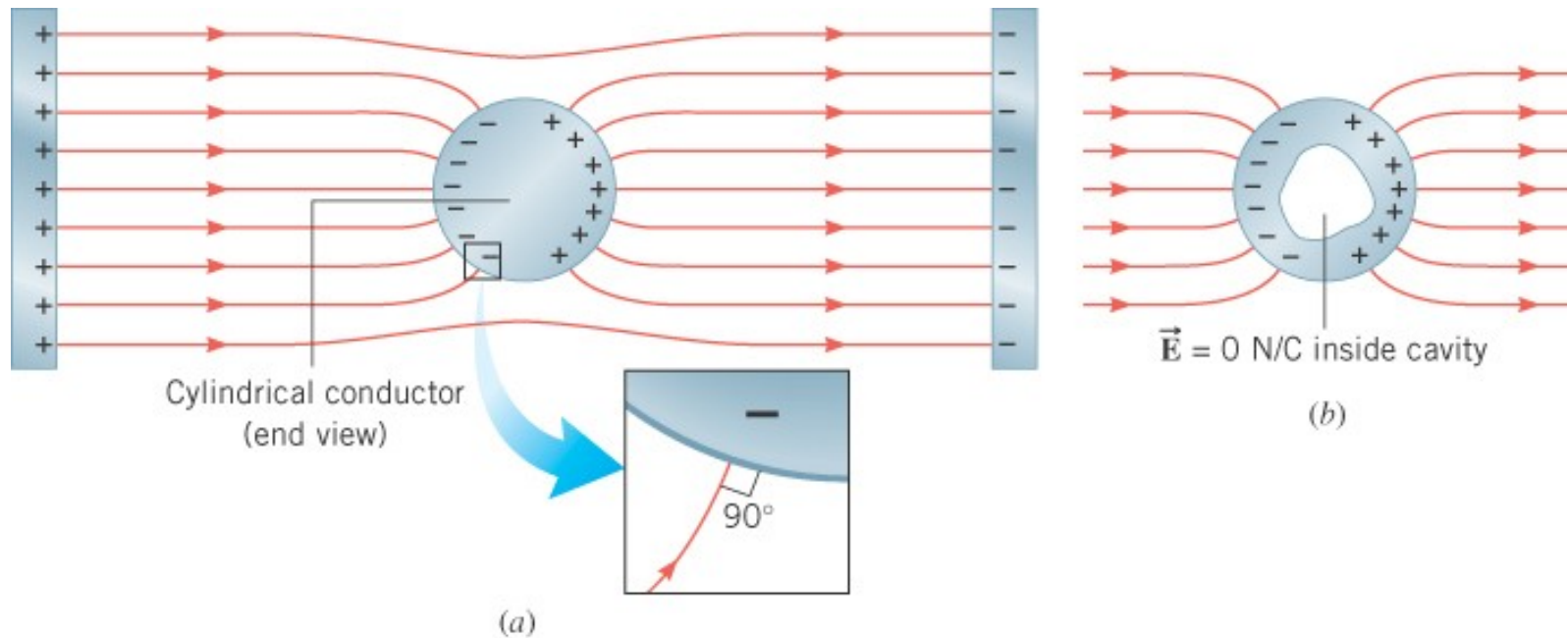
USB cable



Van graaf generator - no charge inside – charges are brought outside



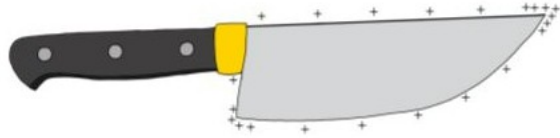
18.8 The Electric Field Inside a Conductor: Shielding



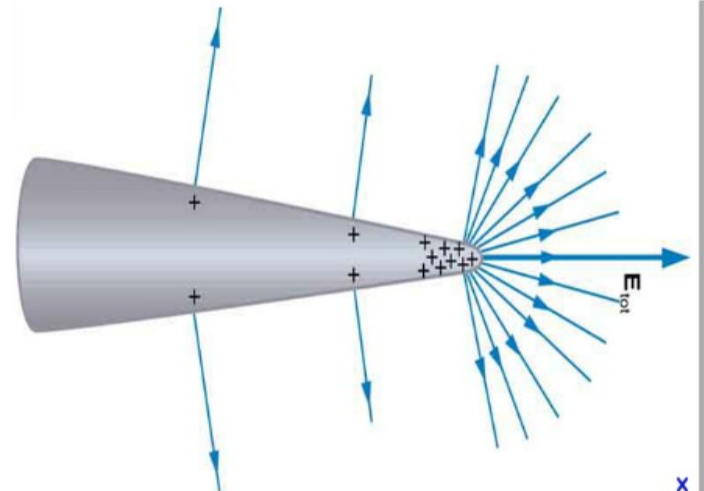
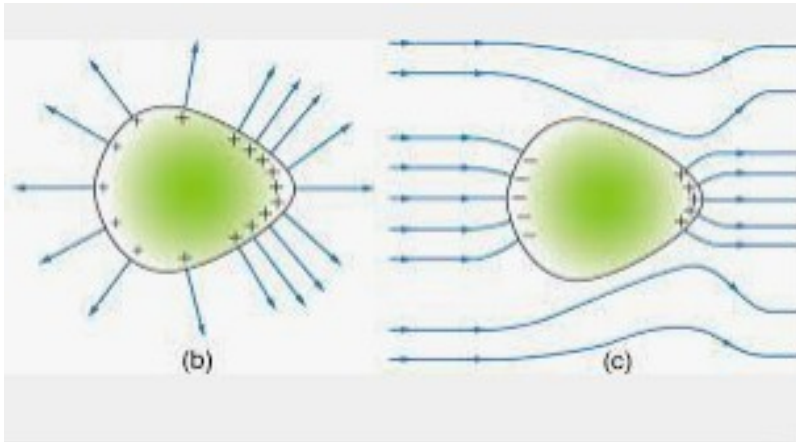
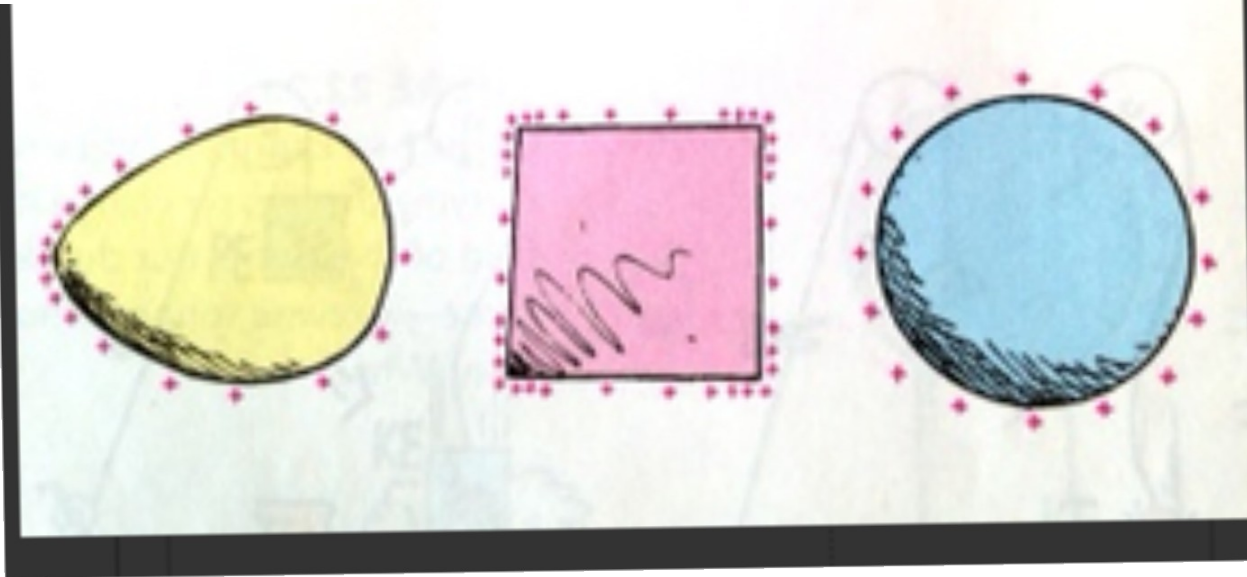
The electric field just outside the surface of a conductor is perpendicular to the surface at equilibrium under electrostatic conditions.

And it is constant at distance small relative to the size of the conductor

$$E = \frac{\sigma}{2\epsilon_0}$$



More charges on small curvature



Gauss' Law Symmetry

How to apply Gauss's law?

Use close surface through which the Electric field is constant in magnitude.

Use symmetry.

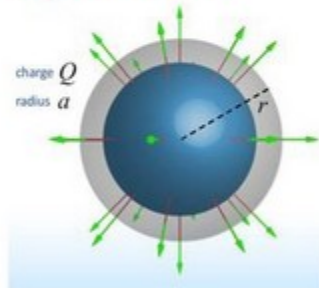
$$\int \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

ALWAYS TRUE!

In cases with symmetry can pull E outside and get

$$E = \frac{Q_{enc}}{A\epsilon_0}$$

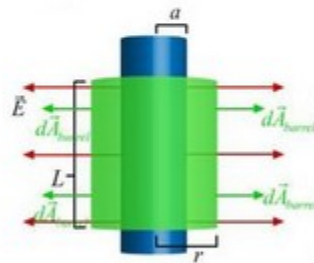
Spherical



$$A = 4\pi r^2$$

$$E = \frac{Q_{enc}}{4\pi r^2 \epsilon_0}$$

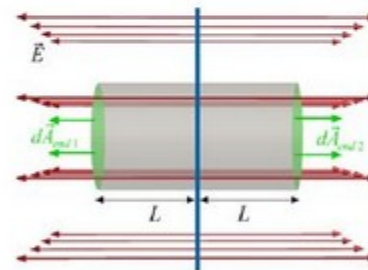
Cylindrical



$$A = 2\pi rL$$

$$E = \frac{\lambda}{2\pi r \epsilon_0}$$

Planar



$$A = 2\pi r^2$$

$$E = \frac{\sigma}{2\epsilon_0}$$

1- Spherical symmetry

(the Gaussian surface is a sphere)

This is not a conductor. This is a nonconducting sphere.

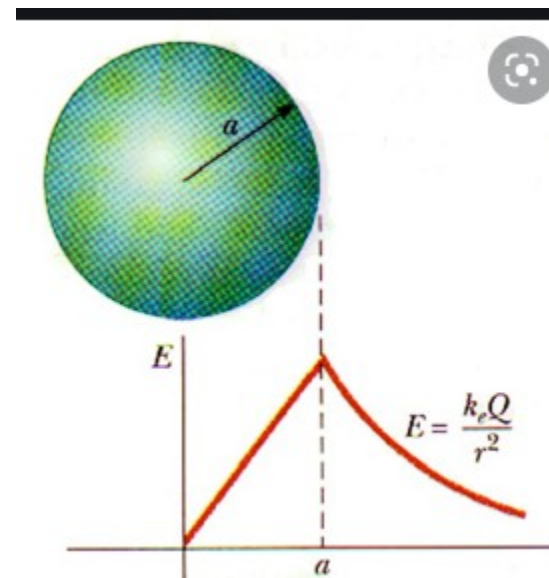
The charges are stuck inside uniformly. (ideal situation)

With a conductor, charges are only spread on the outside.

- A solid sphere of radius R has a total charge $+Q$ uniformly distributed over its volume. The electric field at distance $r < R$ is

- A. proportional to $1/r$.
- B. proportional to $1/r^2$.
- C. proportional to r .
- D. independent of r .

Solution next slide

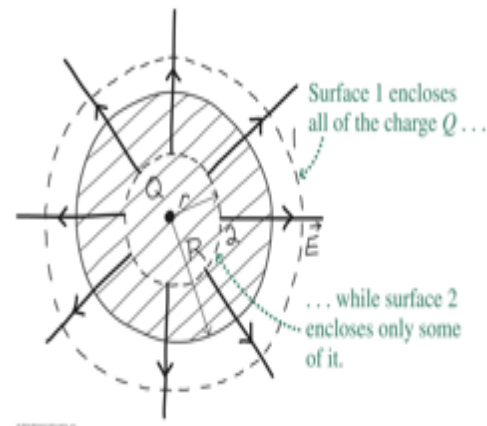


The Field of a Uniformly Charged Sphere

- **INTERPRET:** The situation has spherical symmetry.
- **DEVELOP:** Appropriate Gaussian surfaces are spheres.
- **EVALUATE:**

- The flux becomes $\Phi = \oint \vec{E} \cdot d\vec{A} = 4\pi r^2 E$
- Outside the sphere, the enclosed charge is the total charge Q . Then

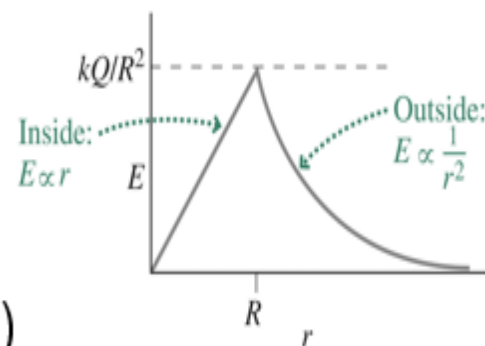
$$4\pi r^2 E = Q/\epsilon_0, \text{ so } E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{kQ}{r^2}$$



- The field outside a spherical charge distribution is identical to that of a point charge.

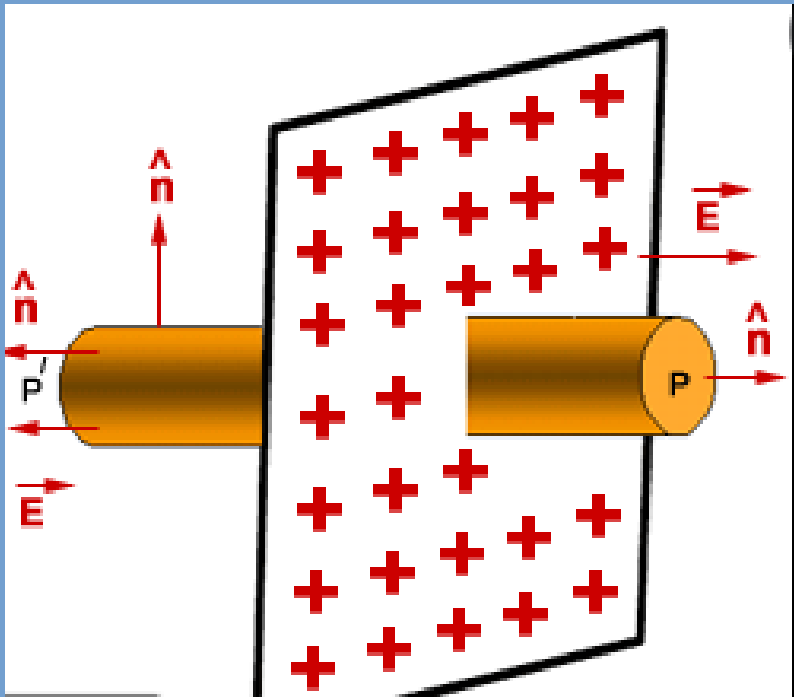
- Inside the sphere, the enclosed charge is proportional to the volume enclosed.

- So $q_{\text{enclosed}} = (r^3/R^3)Q$, $4\pi r^2 E = (r^3/R^3)$
- $$E = \frac{Qr}{4\pi\epsilon_0 R^3}$$



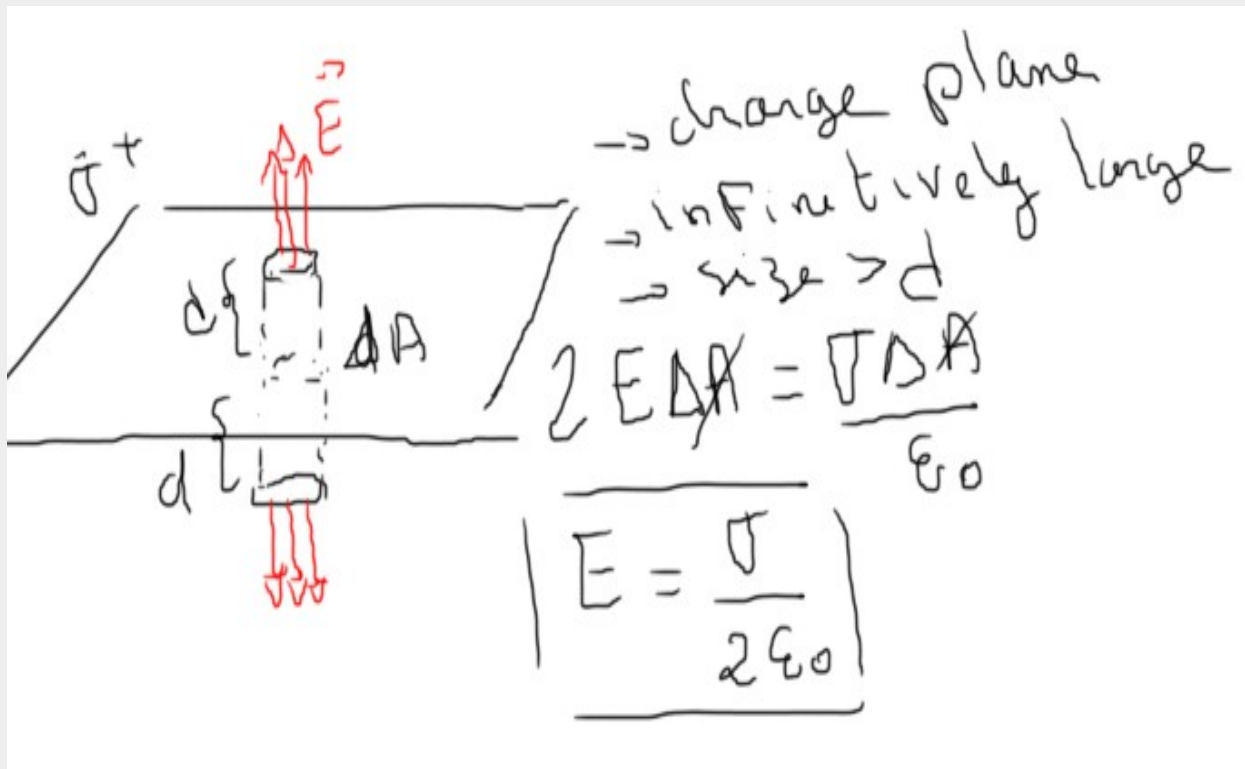
2- PLanar symmetry

(the Gaussian surface is a closed cylinder here)



→ E does not depend
On the distance!
→ only true for
small Distance d
→ show how to
Use Gauss law
→ see how
The components
Along plane
Cancel because
Its a big plane

→ Plane
symmetry



- A square sheet with charge Q uniformly distributed measures 1 m on each side. Which one of the following expressions would you use to approximate the electric field strength at a distance of 1 cm somewhere near the center of the sheet?

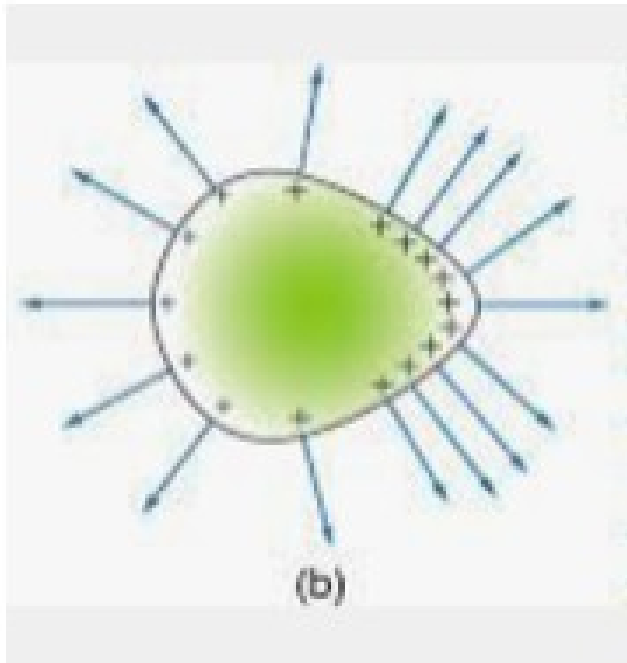
A. $E = \frac{kQ}{r}$

B. $E = \frac{kQ}{r^2}$

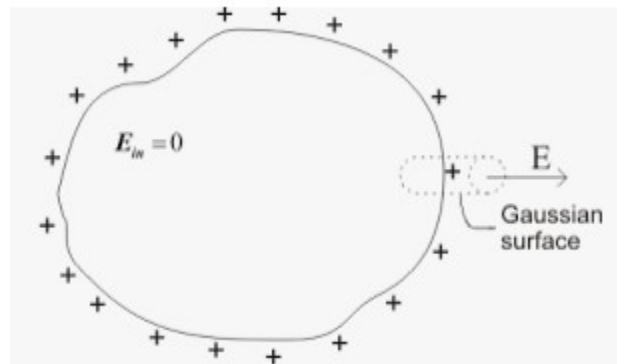
C. $E = \sigma / 2\epsilon_0$

D. $E = \sigma / 4\epsilon_0$

The charge density is not uniform
But at a given point you can show that
The electric field at the surface is:
 σ/ϵ_0

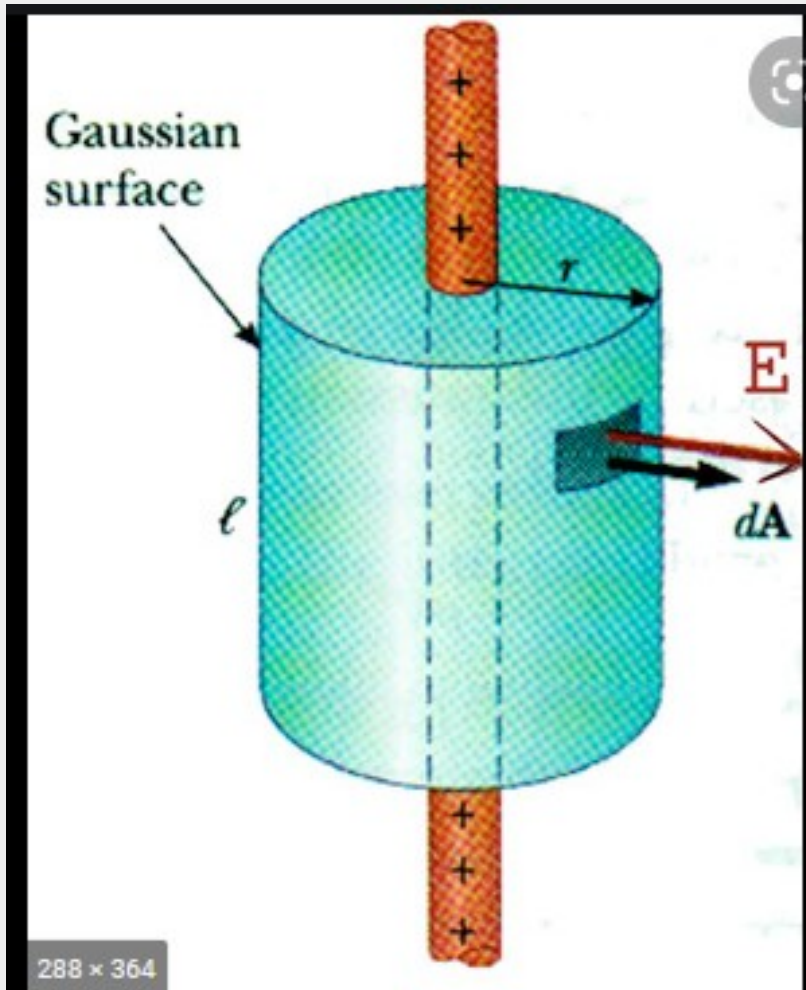


Use a cylinder (closed) so show that.
Note: the electric field is always perpendicular
To surface. Otherwise, the component along
The surface will make the charges move.



2- linear symmetry

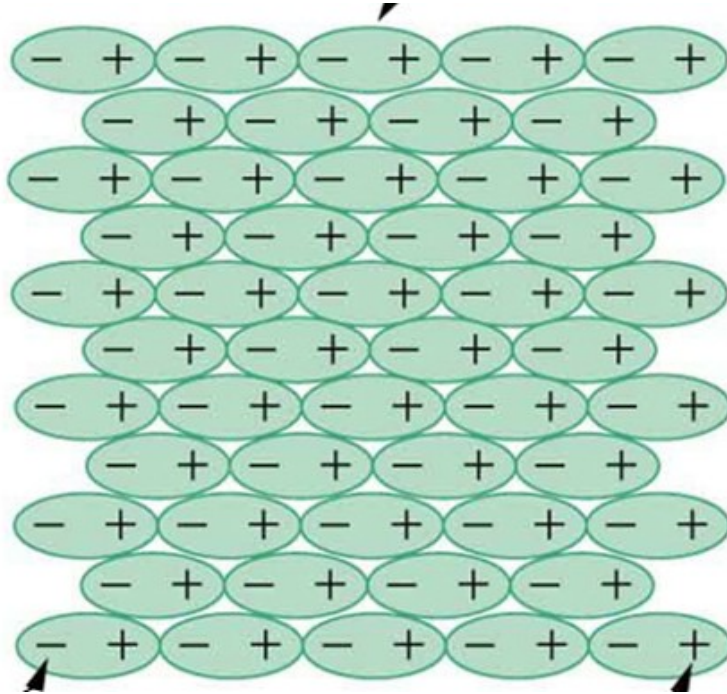
(the Gaussian surface is a closed cylinder here)



$$A = 2\pi rL$$
$$E = \frac{\lambda}{2\pi r\epsilon_0}$$

We can create permanent dipoles

An electret is a dielectric material that has a quasi-permanent electrical charge. Electrets are similar to permanent magnets and can be used to create a permanent electric field. 

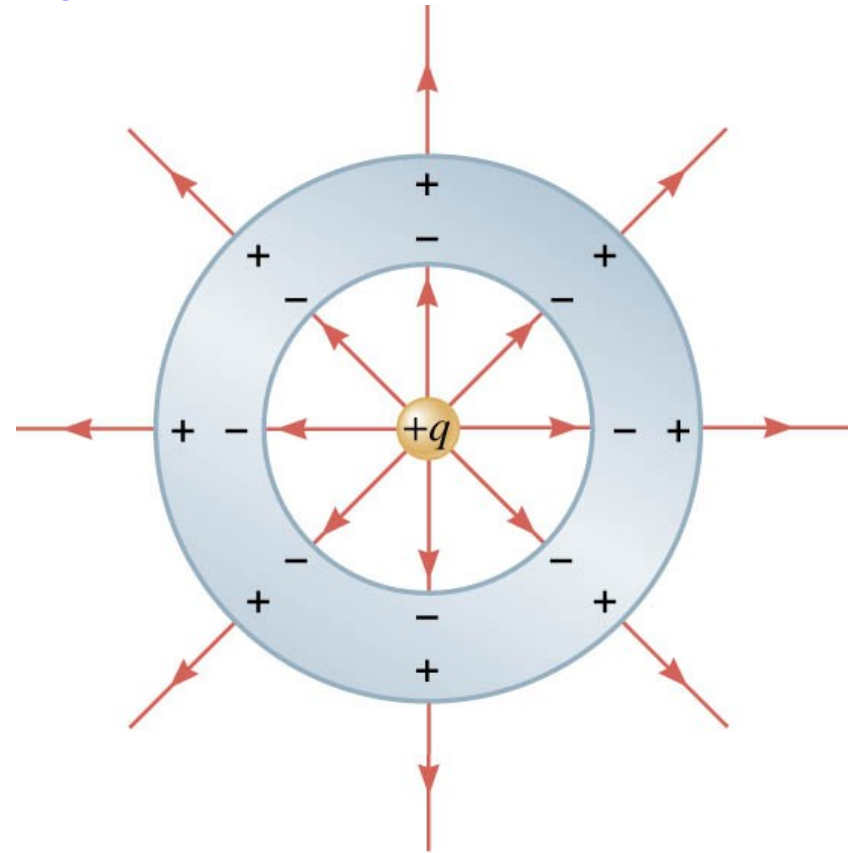


18.8 The Electric Field Inside a Conductor: Shielding

Conceptual Example 14 A Conductor in an Electric Field

A charge is suspended at the center of a hollow, electrically neutral, spherical conductor. Show that this charge induces

- (a) a charge of $-q$ on the interior surface and
- (b) a charge of $+q$ on the exterior surface of the conductor.

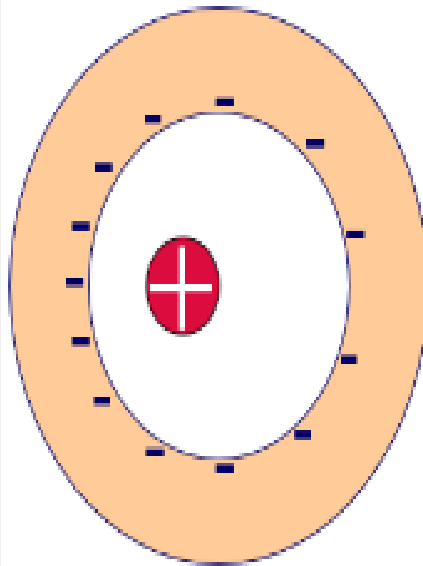


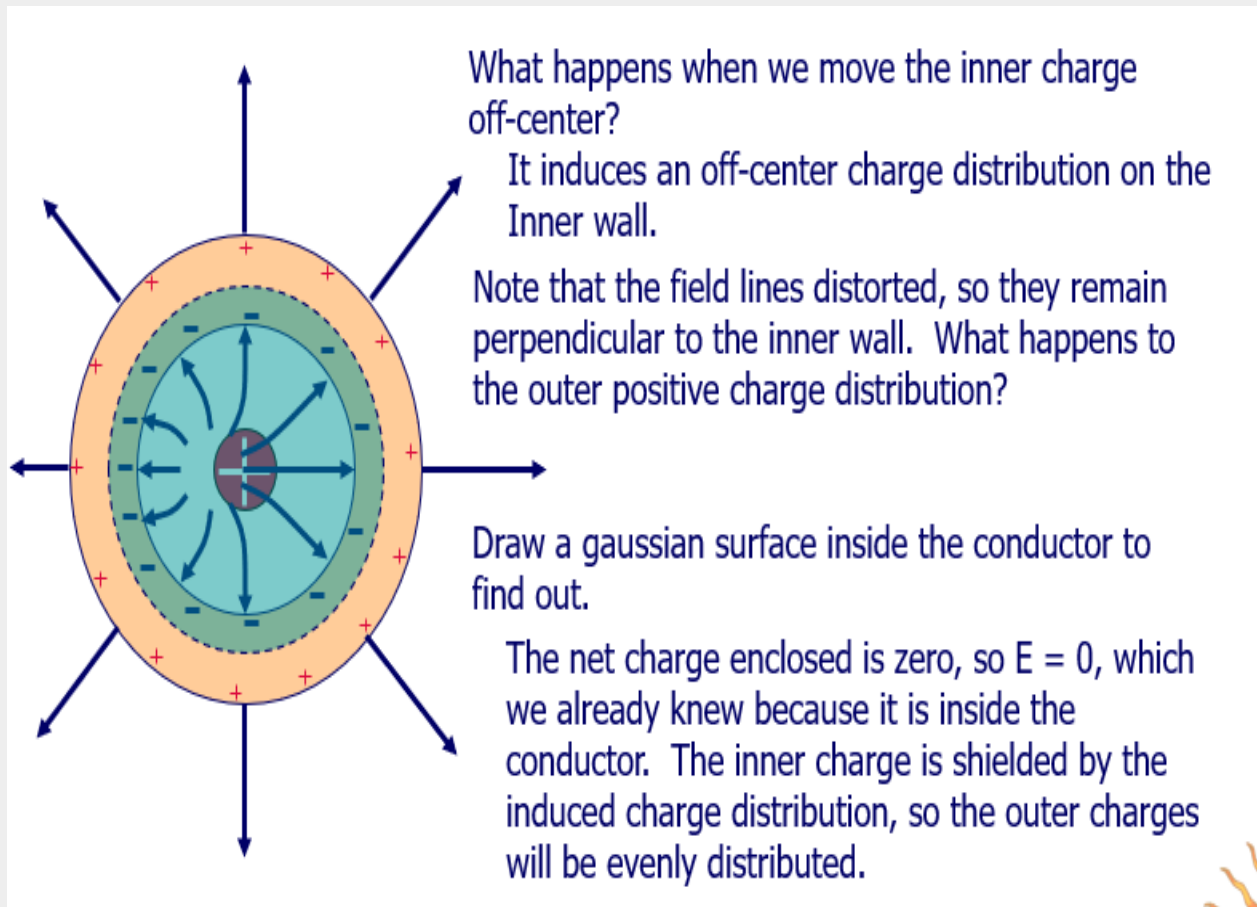
Discuss. Different configuration

E Field of Charge In Conductor

What happens when we move the inner charge off-center?

It induces an off-center charge distribution on the Inner wall.

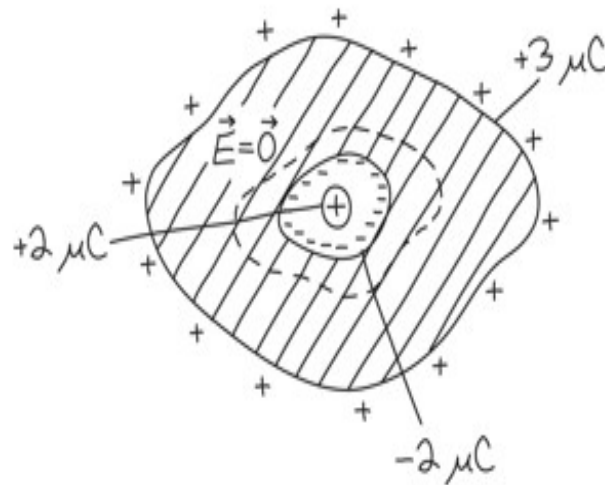




- the outside world does not know what's going on inside.
- coaxial cable
- Faraday's cage

Charged Conductors

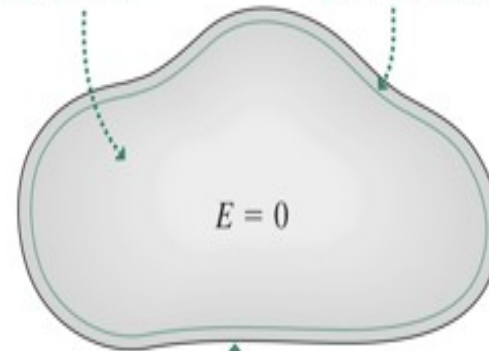
- Gauss's law requires that any free charge on a conductor reside on the conductor surface.



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There's no electric field inside the conductor . . .

. . . so there's no flux Φ through this Gaussian surface.



Because Gauss's law says $\Phi \propto q_{\text{enclosed}}$, all excess charge resides on the conductor surface.

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- When charge resides inside a hollow, charged conductor, then there may be charge on the inside surface of the conductor.

This charged conductor (shaded) carries a net charge of $1\ \mu\text{C}$. There's a $2\text{-}\mu\text{C}$ point charge within a hollow cavity in the conductor. Notice how the charge redistributes itself to be consistent with Gauss's law.

- A conductor carries a net charge of $+Q$. A cavity within the conductor contains a point charge of $-Q$. What is the charge on the outer surface of the conductor in electrostatic equilibrium?

- A. $-Q$
- B. 0
- C. $+Q$
- D. $+2Q$

Try →

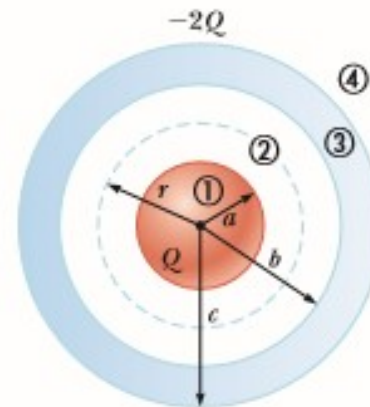
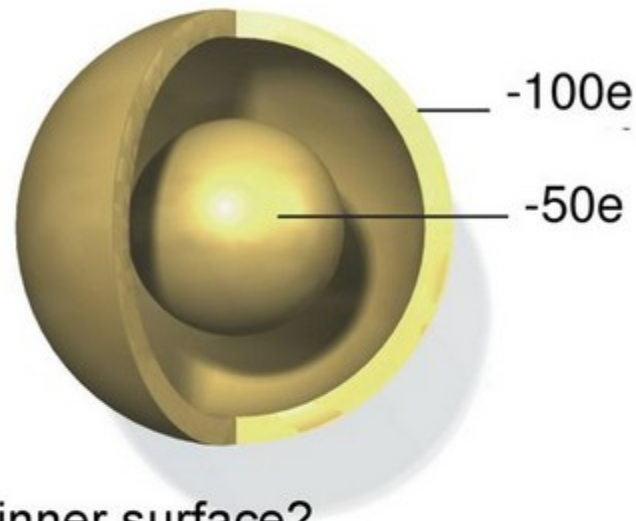


Figure 24.19 (Example 24.7) An insulating sphere of radius a and carrying a charge Q surrounded by a conducting spherical shell carrying a charge $-2Q$.

Example:

A ball of charge $-50e$ lies at the center of a hollow spherical metal shell that has a net charge $-100e$.



(1) What is the charge on the shell's inner surface?

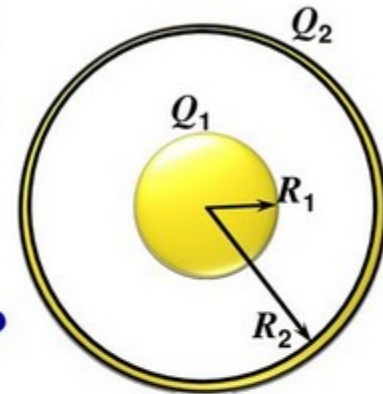
(a) $-50e$

(b) 0

(c) $+50e$

What about the shell?

- A solid conducting sphere is concentric with a thin conducting shell, as shown.
- The inner sphere carries a charge Q_1 , and the spherical shell carries a net charge Q_2 , such that $Q_2 = -3Q_1$.



How is the charge distributed on the sphere?

- (A) There is no charge on the sphere.
- (B) The charge is uniformly distributed on the outside surface of the sphere.
- (C) The charge is uniformly distributed throughout the sphere.

What about the shell?